Engineering Graphics

A standards-first drawing handbook for converting engineering ideas into clear, measurable and manufacture-ready views.

Course 1003 **Semester I** **Drawing** **45 contact hours** **Malayalam support**

Course code	**1003**
Revision	**2026**
Programme	**Common to 20 programmes**
Semester	**I**
Type	**Drawing**
Teaching scheme	**0:0:3:0**
Credits	**1.5**
CIE / ESE	**40 / 60**

Official Source Declaration and Verified Inconsistencies

The curriculum source for this handbook is the supplied SITTTR Kerala **Diploma Curriculum Revision 2026** syllabus PDF for **Course 1003 — Engineering Graphics**. No Revision 2021 course content has been merged. Explanations, diagrams and practice problems below are instructional expansions of the listed 2026 topics, not replacements for official wording.

Verified identity

- Course title: **Engineering Graphics**
- Course code: **1003**
- Semester: **I**
- Course type: **Drawing**
- Contact hours: **45**
- Credits: **1.5**
- Prerequisite topic: **Basic Geometry**

Fields not specified

The supplied official PDF does not specify a single named department because the course is common to multiple programmes. It also does not specify drawing-board size, exact sheet size for every exercise, instrument brands, line-width values, lettering height for every sheet, or laboratory-room equipment quantities. These must follow current institution/BIS practice and instructor direction.

Official-source inconsistency 1 — fourth teaching-scheme symbol: Page 1 labels the scheme as **L:T:P:R**, while page 2 labels the same four values as **L:T:P:S** and defines S as Project. Both tables show **0:0:3:0**. This handbook displays the numeric scheme exactly and describes it as 3 drawing/practical periods per week without redefining the disputed fourth symbol.

Official-source inconsistency 2 — delivery classification: The course is classified as Drawing and the weekly scheme gives L=0, T=0, P=3. However, every detailed-syllabus row is marked **L** in a column titled “Mode of delivery (L/T).” This handbook preserves the official topic hours and treats them as drawing-studio instructional hours.

Official-source inconsistency 3 — assessment heading: The teaching-and-assessment table and page-4 heading place CIE 40 and ESE 60 under “Theory/Assessment Methodology — Theory,” while the course and question-paper patterns are explicitly labelled Drawing. The handbook uses the official marks and drawing question-paper pattern and flags the heading mismatch rather than silently correcting it.

Official-source inconsistency 4 — prerequisite table: “Basic Geometry” appears under Topic; the Course code and Course Title cells are blank; “Secondary Education” appears under Semester. This handbook states only the unambiguous prerequisite: Basic Geometry at secondary-education level.

Official-source formatting ambiguity 5 — CIE conversion: The page-4 merged-cell layout clearly gives the final CIE grouping as Attendance 5 + Self-learning activities 15 + Series examinations 20 = 40, but it does not give an unambiguous separate conversion formula for each CA item. No unsupported conversion formula is invented here.

COURSE DASHBOARD

Teaching, outcomes and assessment at a glance

45 h

Official contact hours

22.5 h

Official self-learning total

1.5

Credits

4

Course Outcomes and modules

Official teaching and examination scheme									
L	T	P	Fourth field	Self-learning/week	Total hours	Credits	CIE	ESE	Total
0	0	3	0 (R on page 1; S on page 2)	1.5 h	45 h	1.5	40	60	100

Programmes listed in the official course description

Automobile Engineering; Civil Engineering; Chemical Engineering; Civil Engineering & Planning; Civil & Rural Engineering; Civil & Environmental Engineering; Electrical & Electronics Engineering; Electrical Engineering; Electrical Engineering & Electric Vehicles Technology; Fire Technology and Safety; Food Processing Technology; Mechanical Engineering (Automobile Engg.); Mechatronics; Mechanical Engineering; Polymer Technology; Printing Technology; Renewable Energy; Civil Engineering (Construction Technology); Tool & Die Engineering; Wood and Paper Technology.

Continuous Internal Evaluation — 40

- Attendance: 5 marks
- Self-learning activities across Modules I–IV: 15 marks
- Two series-exam components together: 20 marks
- Official series examination duration: 2 hours; raw mark shown: 50

End Semester Examination — 60

- Duration: 2.5 hours
- Each module contributes 15 marks
- Part A: one out of two 5-mark questions per module
- Part B: one 10-mark set with choice per module

Course Outcomes — official wording

CO1 · Understand

Illustrate the basic elements of drawing.

Student meaning: select correct lines, instruments, standards, dimensions and polygon construction.

CO2 · Apply

Construct geometrical figures and illustrate development of surfaces.

Student meaning: execute the prescribed conics, involutes and developments accurately.

CO3 · Understand

Interpret projections of points and lines.

Student meaning: convert 3D location statements into front and top views.

CO4 · Apply

Develop orthographic projections and sectional views of objects.

Student meaning: correlate views, reveal internal details and follow first-angle conventions.

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	–	–	2	–	–	–	1	–	–
CO2	3	2	2	–	2	–	–	–	1	–	–
CO3	3	3	2	1	2	–	–	–	1	–	–
CO4	3	3	3	2	2	–	–	–	1	–	–

Legend: 1 Low, 2 Medium, 3 High, “–” No correlation.

STUDY METHOD

How to use this handbook

1 · Understand

Read the concept and vocabulary first. Do not start instrument construction without understanding what each line represents.

2 · Construct

Follow the numbered procedure using thin construction lines, verify geometry, then darken only the final result.

3 · Review

Use Malayalam support, solved demonstrations, error warnings, expected questions and the final quick-revision checklist.

> **Malayalam support:** Drawing [illegible] “steps [illegible]” [illegible] [illegible]. [illegible] construction line [illegible] [illegible] [illegible] [illegible]. [illegible] thin line, [illegible] verification, [illegible] [illegible] final dark line [illegible] [illegible] [illegible].

OFFICIAL DIRECTION

Course objectives

1. To familiarize the language of graphics which is used to express ideas, convey instructions while carrying out engineering jobs.
2. To familiarize drafting and sketching skills, to know the applications of drawing equipment’s, and get familiarize with Indian Standards related to engineering drawings, dimensioning styles etc.
3. To provide skills to translate ideas into sketches and to draw and read various engineering curves, conic sections and the generation of 3D solid surfaces into plane areas.
4. To apply basic skills to develop projection of points and lines.
5. To familiarize the knowledge of orthographic views and sectional views of objects in engineering applications.

CURRICULUM COVERAGE MAP

Every official topic mapped to a handbook chapter

Module-hour and CO coverage

Module	Official topics	Hours	CO	Cognitive level	Handbook section
I — Introduction	Basic terminology; dimensioning methods and scales for study; regular pentagon, hexagon, heptagon, octagon with side given	9	CO1	Remember / Understand	Module I
II — Basic Figures	Ellipse by rectangular and concentric-circle methods; parabola by tangent method; involute of circle and rectangle; developments of rectangular prism, cylinder and 90° two-piece symmetric elbow	12	CO2	Understand / Apply	Module II
III — Basic Projections	Projection theory; first- and third-angle projection; points in four quadrants; lines in first quadrant, inclined to one plane only	9	CO3	Understand	Module III
IV — Projections–3D	Orthographic and isometric projections; orthographic projections of simple objects; full sectional views of simple objects	15	CO4	Understand / Apply	Module IV

Learning roadmap

1. Standards
Build the drawing language. →

2. Geometry
Construct curves and unfold surfaces. →

3. Projection
Locate points and lines. →

4. Views
Describe and section objects. ✓

MODULE I · 9 INSTRUCTIONAL HOURS

Introduction — terminology, dimensioning and regular polygons

CO1 | Remember / Understand | 3 + 3 + 3 h

Engineering graphics is controlled communication. A technically correct drawing tells another person what exists, where it is, and how large it is—without depending on spoken language.

Malayalam support: Engineering Graphics □□□ picture drawing □□□□; standardized technical communication □□□. line type, symbol, view arrangement, dimension □□□□□□□□□ consistency □□□□□□□□□□ drawing □□□□□□□□ manufacture □□□□□□□□□□□□.

Module objectives

- Identify drawing media, instruments and sheet layout.
- Use standard line meanings and lettering habits.
- Apply sound dimensioning principles.
- Recognise scales for study purpose.
- Construct four prescribed regular polygons.

Quality target

- Thin construction lines remain distinguishable.
- Final outlines are clean and continuous.
- Dimensions are complete but not duplicated.
- Polygon sides verify equal before darkening.

Industrial link

Production, inspection, CAD, fabrication, construction and maintenance personnel must interpret the same geometry. Standards reduce ambiguity and rework.

1.1 Basic terminology, drawing materials and standards — 3 h

Core vocabulary

Drawing

A graphical specification of shape, size, location and other technical requirements.

View

The appearance of an object obtained from a chosen direction of sight.

Projection

The process of transferring object information to a plane using imaginary projectors.

Construction line

A very light working line used to locate geometry; it is not the finished object boundary.

Object line

A prominent continuous line representing visible edges and outlines.

Title block

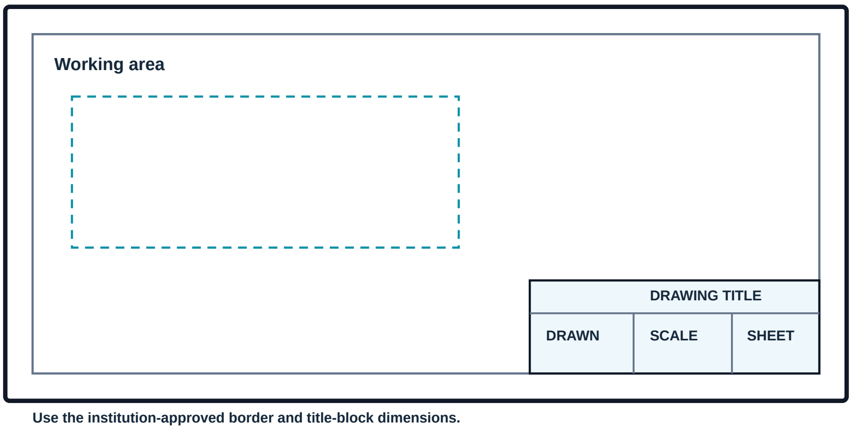
The controlled area containing drawing identification, title, scale, sheet and approval information.

Basic instruments and correct use

Instrument / material	Purpose	Good practice	Typical error
Drawing board	Flat reference surface	Keep clean, dry and free from dents	Using a damaged edge as a reference
Mini drafter / T-square	Horizontal and controlled angular lines	Lock gently and verify alignment	Forcing a loose or misaligned head
Set squares	Vertical, 30°, 45°, 60°, 90° lines	Hold contact edges together	Drawing along a chipped edge
Compass	Circles and arcs	Keep needle and lead at comparable reach; rotate smoothly	Changing radius during rotation
Divider	Transfer and divide distances	Use light pricks, not deep holes	Measuring by scale after equal-step division has begun
Pencils	Construction and finished lines	Use a sharp point and pressure suited to line function	Trying to obtain thick lines with a blunt uncontrolled point
Scale	Set off dimensions	Read the correct graduation and avoid parallax	Using it as a cutting or ruling edge when unsuitable
French curve / templates	Smooth non-circular curves and repeated shapes	Match several adjacent points before drawing	Joining curve points with short straight chords

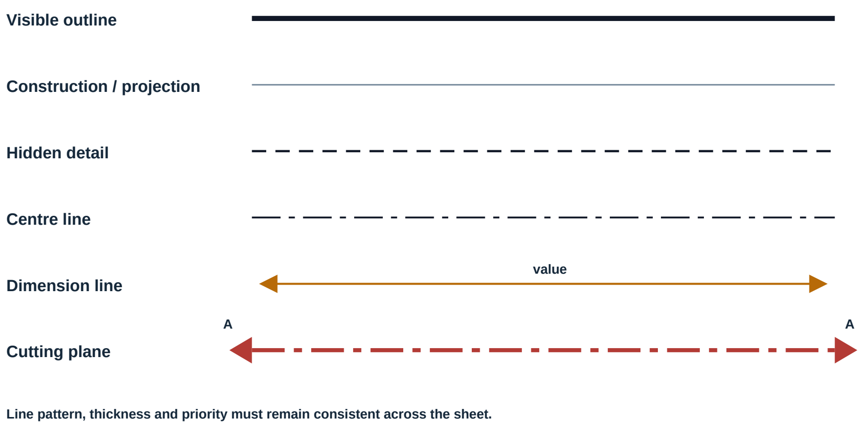
Sheet sizes, layout and title block

The official self-learning activity requires familiarisation with drawing-sheet sizes, board types, sheet layout and title block. ISO/BIS A-series sheets retain a constant aspect ratio; the selected sheet and border/layout must follow the current institutional drawing standard. A title block normally identifies the drawing title, drawing number, scale, sheet number, drafter, checker and date. Do not invent project-control fields when the institution supplies a template.



Typical layout concept. Exact border offsets and title-block fields must follow the current institution/BIS template.

Line types and their meaning



Line function matters more than decoration. Never use the same visual weight for every line.

Malayalam support: Visible edge thick continuous line □□□, hidden edge dashed line □□□, axis/centre chain line □□□ □□□□□□□□□□□□. Construction line □□□□ light □□□□□□□□□□. □□□□□ lines-□□□ □□□ thickness □□□□□□□□□ drawing □□□□□□□□□ □□□□□□□□□□□□□□□.

Priority at coincidence: when different line types overlap, the line representing visible shape normally takes priority over hidden and centre information. Avoid drawing redundant lines on top of each other.

1.2 Principles of dimensioning and scales — 3 h

What a dimension must communicate

A dimension specifies a measurable requirement. A complete drawing must state enough sizes and locations to define the object without making the reader calculate avoidable values or measure the picture. The numerical value—not the apparent drawn length—controls the part.

Sound dimensioning rules

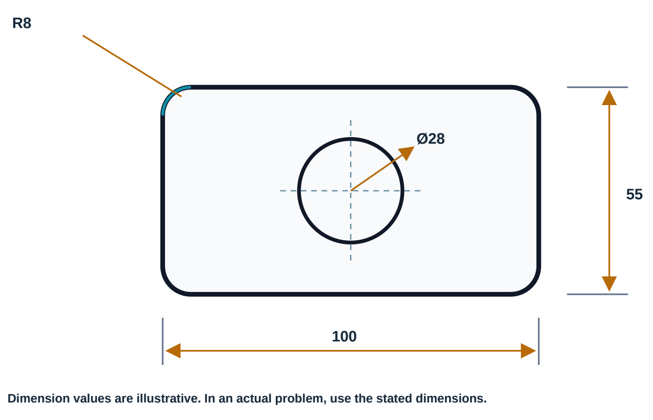
- Give each necessary dimension once.
- Place dimensions outside the view where practical.
- Dimension from functional or stable reference features.
- Avoid dimensioning to hidden lines when a visible or sectional view is available.
- Do not cross dimension lines unnecessarily.
- Keep arrowheads, text orientation and spacing uniform.
- Use symbols such as Ø for diameter and R for radius correctly.
- State the unit convention in the drawing or title block; do not mix units silently.

Aligned and unidirectional systems

Aligned: dimension numerals align with their dimension line and are read from the bottom or right side of the sheet.

Unidirectional: all numerals are horizontal and are read from the bottom of the sheet.

Do not mix systems within one drawing unless an approved standard/template explicitly requires it.



Extension lines begin slightly away from the object. Dimension lines terminate with consistent arrowheads and carry a clear value or symbol.

Dimensioning checklist before submission

1. Can the object be made without measuring the drawing?
2. Is every dimension necessary, and has any value been repeated?
3. Are diameters, radii and angles identified with correct symbols?
4. Do extension lines extend slightly beyond the dimension line?
5. Are numerals readable in one consistent dimensioning system?
6. Are centre lines present for circles and symmetry?

Scales — study purpose only

The syllabus includes Plain, Diagonal and Vernier scales for study purpose. A scale relates a represented length to the actual length. The **Representative Fraction (RF)** is dimensionless when both lengths are expressed in the same unit.

RF = length on drawing ÷ corresponding actual length
Full size: RF = 1; reducing scale: RF < 1; enlarging scale: RF > 1.

Plain scale

Reads a main unit and one smaller subdivision. Example: metres and decimetres.

Diagonal scale

Uses similar triangles to obtain a second finer subdivision beyond the plain-scale reading.

Vernier scale

Uses the difference between main-scale and vernier divisions to read a smaller least count.

Worked study example — scale length

Problem: A 12 m long object is represented at RF 1:50. Find the required drawing length.

1. Actual length = 12 m = 12,000 mm.
2. Drawing length = RF × actual length.
3. Drawing length = (1/50) × 12,000 = 240 mm.

Answer: 240 mm. Always convert actual and drawing lengths to the same unit before applying RF.

Malayalam support: Scale □□□□□□ ruler □□□□; drawing length-□□□ actual length-□□□ □□□□□□□□□ ratio □□□. RF calculate □□□□□□□□□□□ numerator, denominator □□□□□□ same unit □□□□□□.

1.3 Construction of regular polygons — 3 h

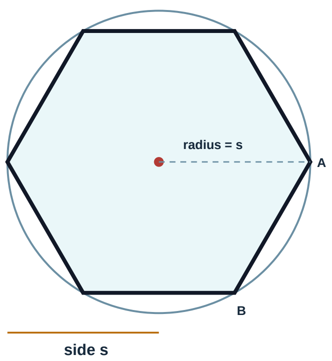
A regular polygon has equal sides and equal interior angles. The official syllabus requires construction of a **pentagon, hexagon, heptagon and octagon** when the side length is given, using any suitable method for each type.

Interior angle of a regular n-sided polygon = (n − 2) × 180° / n
Exterior (turning) angle = 360° / n

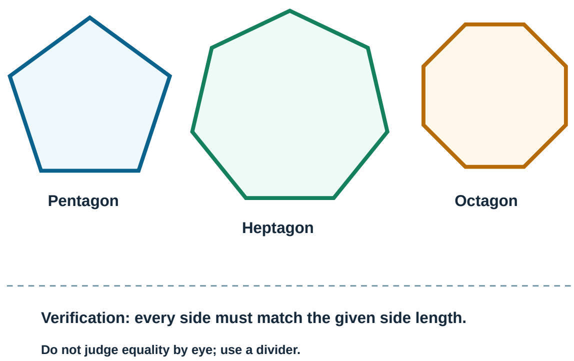
Polygon	n	Interior angle	Exterior angle	Construction note
Pentagon	5	108°	72°	Use an exact pentagon construction or a verified general polygon method.
Hexagon	6	120°	60°	For a regular hexagon, side equals circumradius.
Heptagon	7	≈128.57°	≈51.43°	A suitable general/approved construction is acceptable; verify all sides.
Octagon	8	135°	45°	Can be developed from a square/circumscribed-circle construction.

Reliable construction workflow

1. **Set the given side.** Draw AB exactly and leave enough surrounding space.
2. **Select the approved method.** Use the method taught by the instructor for that polygon; do not combine unrelated constructions halfway.
3. **Locate centre or successive vertices.** Keep arcs and projectors light but visible.
4. **Join in cyclic order.** Avoid crossing the vertex sequence.
5. **Verify.** Check all side lengths with a divider and inspect symmetry before darkening.



Hexagon: set compass radius equal to the given side and step the same chord six times around the circle.



Regular means equal sides and equal angles. A shape that merely looks symmetric is not enough.

Malayalam support: Regular polygon-□ □□□□□ sides-□□ equal, □□□□□ internal angles-□□ equal □□□. Final line dark □□□□□□□□□□□□□ □□□□□□ divider □□□□□□□□□□ □□□ side-□□ check □□□□□□□.

Common failure: students often preserve equal-looking angles but lose the given side length. The given side is the controlling datum; verify it at every edge.

Module I summary and self-check

One-look summary

- Standards give every line and symbol a shared meaning.
- Construction lines locate; object lines describe.
- Dimension numerals control, not the visual scale of the picture.
- RF is a ratio in common units.
- A regular polygon must pass a divider check.

Self-check exercises

1. Prepare a sheet of six line types and label their use.
2. Correct a drawing containing duplicate and hidden-line dimensions.
3. Calculate drawing length for 8 m at 1:40.
4. Construct the prescribed four polygons for a side selected by the instructor.

Check for Exercise 3: 8 m = 8000 mm; 8000/40 = **200 mm**.

MODULE II · 12 INSTRUCTIONAL HOURS

Basic figures — conic sections, involutes and developments

CO2 | Understand / Apply | 4 + 4 + 4 h

This module converts mathematical geometry into practical construction. Accuracy depends on locating enough points, preserving symmetry and transferring true lengths without cumulative error.

Malayalam support: Curve construction-□ point plotting □□□□□□□□□□; points □□□□□□ straight lines □□□□□□□□□□. Development-□ 3D surface flat sheet □□□ □□□□□□□□□□□□ true length □□□□□□ transfer □□□□□□□.

2.1 Conic sections — ellipse and parabola — 4 h

Conic terminology

A conic section is a curve obtained conceptually by intersecting a right circular cone with a plane. In this syllabus, the required constructions are limited to **ellipse by rectangular method and concentric-circle method**, and **parabola by tangent method**.

Ellipse

A closed curve with two perpendicular axes. The longer is the major axis and the shorter is the minor axis. Their intersection is the centre.

Parabola

An open symmetric curve. In focus-directrix language, every point is equidistant from a fixed focus and a fixed directrix.

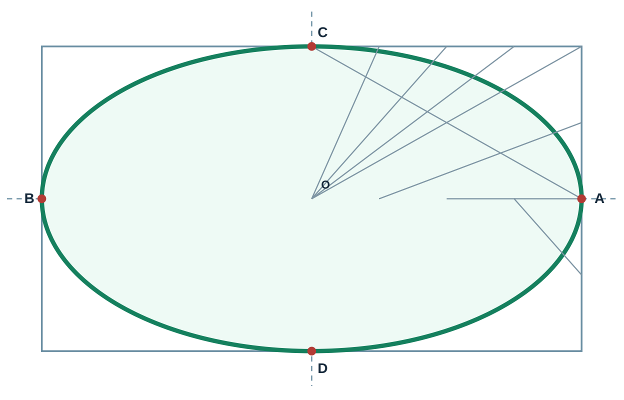
Tangent

A straight line that touches a curve locally. A tangent-envelope construction builds a curve as the smooth envelope of many tangents.

Ellipse — rectangular (oblong) method

Use this method when major and minor axes are given. The rectangle acts as a coordinate frame; corresponding divisions generate points of the ellipse.

1. Draw major axis AB and minor axis CD perpendicular at centre O. Through A, B, C and D, complete the bounding rectangle.
2. Work first in one quadrant. Divide the semi-major axis and the adjacent half-side of the rectangle into the same number of equal parts.
3. Join the half-side divisions to one end of the major axis, and the semi-major divisions to the appropriate end of the minor axis, following one consistent numbering direction.
4. The intersections of corresponding construction lines locate points of the ellipse in that quadrant.
5. Repeat or reflect the points about both axes. Draw one smooth curve through A, C, B, D and all intermediate points.

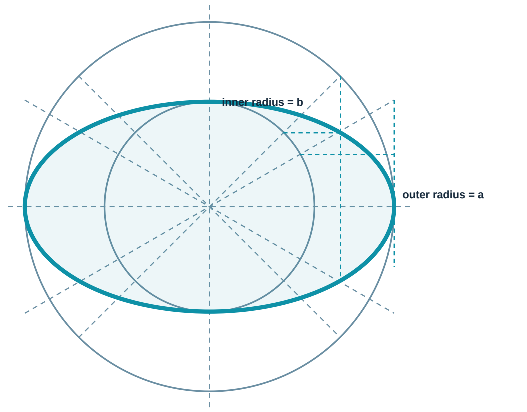


Rectangular method: construction occurs within the bounding rectangle; reflect accurately across both axes.

Ellipse — concentric-circle method

1. Draw the major and minor axes through centre O.

2. With O as centre, draw an outer circle of radius equal to the semi-major axis and an inner circle of radius equal to the semi-minor axis.
3. Divide both circles by the same radial lines into equal angular parts.
4. From each point on the outer circle draw a line parallel to the minor axis; from the matching point on the inner circle draw a line parallel to the major axis.
5. Each pair intersects at an ellipse point. Join all points smoothly.

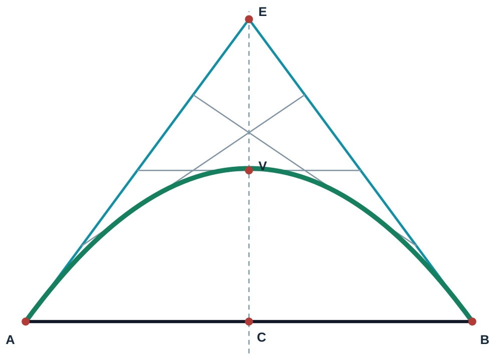


Concentric-circle method. Use the same radial division for both circles; mismatched numbering destroys symmetry.

Parabola — tangent method

The tangent method constructs a family of straight lines whose smooth envelope is the parabola. The exact starting dimensions and labels depend on the problem statement. The following standard workflow uses base AB, axis CV and vertex V.

1. Draw base AB and locate its midpoint C. Draw axis CV perpendicular to AB to the stated vertex V.
2. Extend the axis beyond V to E such that VE = VC. Join E to A and E to B; these act as end tangents.
3. Divide EA and EB into the same number of equal parts, numbered in corresponding opposite order as taught in the construction.
4. Join corresponding division points across the two end tangents. These lines form the tangent family.
5. Draw a smooth envelope through A, V and B, touching the family rather than cutting sharply across it.



The green curve is drawn as the smooth envelope of the tangent family.

Tangent construction concept. Follow the instructor’s exact division numbering; the curve should touch the tangent family smoothly.

Curve quality warning: do not draw a polyline through a few points. Use a French curve or a controlled smooth stroke, match several points at a time, and avoid a kink at the axis.

Malayalam support: Ellipse-□ major/minor axes □□□□□□□□ symmetry □□□□. Parabola tangent method-□ straight tangent lines-□□□□ smooth envelope □□□ final curve; lines-□□ cut □□□□□□ □□□□□□□ zig-zag curve □□□□□□□□□□□.

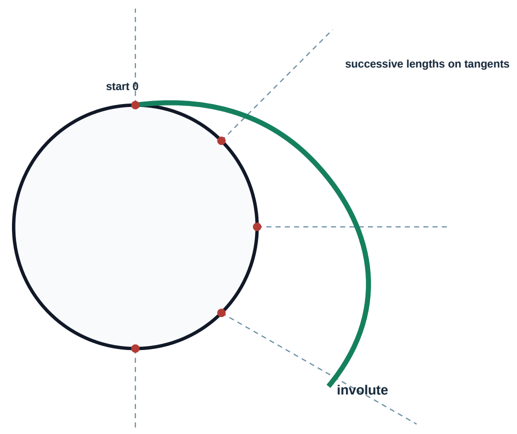
2.2 Miscellaneous curves — involute of circle and rectangle — 4 h

Meaning of an involute

An involute is the locus traced by the end of a taut string as it is unwound from a base curve. The string remains tangent to the base at the current contact point. This geometric idea appears in gear-tooth profiles, cams and winding/unwinding problems.

Involute of a circle

1. Draw the base circle and select a starting point 0. Divide the circumference into equal parts in the direction of unwinding.
2. Draw a tangent at every division point.
3. Let one chord-step approximate one divided arc length. On tangent at point 1 set off one step; at point 2 set off two steps; continue cumulatively.
4. Mark the located points 1′, 2′, 3′ … and draw a smooth curve from 0 through them.

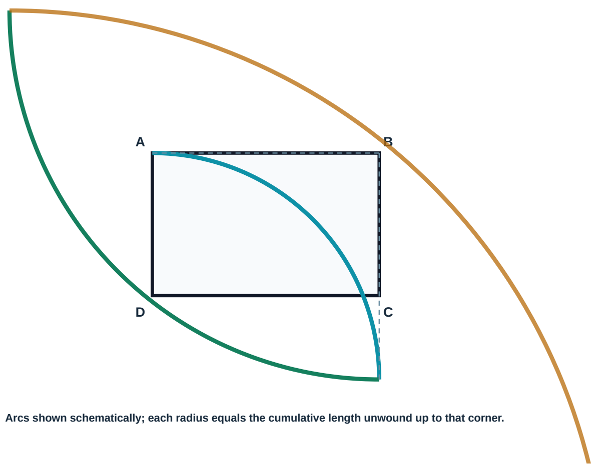


Circle involute: cumulative arc lengths are transferred along successive tangents in the unwinding direction.

Involute of a rectangle

For a polygon, the taut string turns around successive corners. Each new arc is drawn about the next corner, with radius equal to the cumulative length of the already-unwound sides.

1. Draw rectangle ABCD and choose the starting corner and unwinding direction.
2. Extend the appropriate sides in the unwinding direction.
3. With the first corner as centre, draw an arc of radius equal to the first side. For the next arc, use the next corner and cumulative radius of two sides.
4. Continue around the rectangle, increasing the radius by each side length in sequence. Join arcs tangentially.



Rectangle involute. Maintain the specified direction; changing direction midway produces the wrong locus.

Common errors: using chords inconsistently, drawing normals instead of tangents, placing cumulative lengths on the wrong tangent, and changing the unwinding direction.

Malayalam support: Involute □□□□□□ tight string base curve-□ □□□□□□ unwind □□□□□□□□□□□ string end trace □□□□□□□□□ path □□□. □□□ stage-□□□ string base-□□□□ tangent □□□□□□□□□□; distance cumulative □□□ □□□□□□□.

2.3 Development of surfaces — prism, cylinder and 90° elbow — 4 h

Purpose and principle

Development is the unrolled flat pattern of a three-dimensional surface. Sheet-metal fabrication uses developments to cut material before bending or joining. A correct development preserves true lengths and adjacency of surfaces.

Parallel-line method

Used for prisms and cylinders because their generators are parallel. The lateral surface stretches out as a rectangle or a strip of panels.

True length

A length may be transferred directly only when it appears in true size in the selected view or has been found by an approved true-length construction.

Closed surface

“All sides closed” means the lateral development must be accompanied by the required end faces. Joining allowances are added only when the problem/instructor specifies them.

Closed rectangular prism

1. Draw the required views and identify base sides in sequence, for example a, b, a, b.
2. Draw a stretch-out line equal to the perimeter 2(a+b), divided into the same side sequence.
3. At division points draw parallel generators equal to the true prism height h.
4. Join the top edge to complete four lateral rectangles. Attach the two end rectangles at suitable matching edges without overlap.
5. Label edges so the relationship between view and development remains traceable.

Lateral stretch-out length = perimeter of base = $2(a + b)$
Lateral area = perimeter × height = $2(a + b)h$

Closed cylinder

1. Draw the circular end and elevation. Divide the circle into equal parts and number generators consistently.

2. Draw the stretch-out length equal to the circumference πD and divide it into the same number of parts.
3. Draw generators equal to cylinder height h and complete the rectangular lateral surface.
4. Add two circular ends of diameter D at suitable locations as required by the closed-surface exercise.

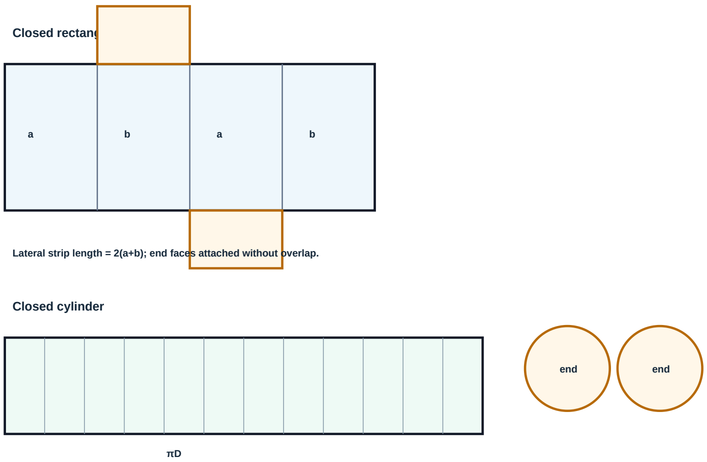
Cylinder lateral stretch-out = πD
Lateral area = πDh
Closed total surface area = $\pi Dh + 2(\pi D^2/4)$

Worked demonstration — cylinder development

Given: cylinder diameter D = 50 mm, height h = 80 mm. **Required:** theoretical lateral rectangle.

1. Stretch-out length = $\pi D = \pi \times 50 \approx 157.1$ mm.
2. Rectangle height = h = 80 mm.
3. Lateral area = $157.1 \times 80 \approx 12{,}568$ mm².

Result: lateral development is approximately 157.1 mm × 80 mm, before any institution-approved seam allowance. For a closed cylinder, add two Ø50 mm end circles.



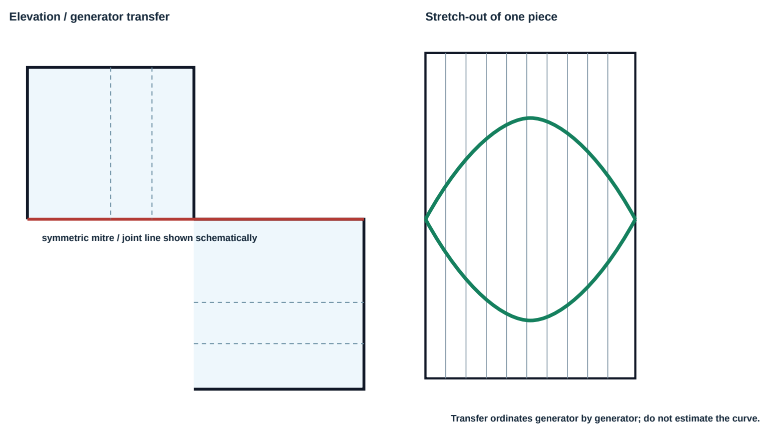
Parallel-line developments. Numbering generators helps transfer intersections and prevents reversed patterns.

90° elbow — two-piece symmetric type

The official exercise is limited to a two-piece symmetric 90° elbow with standard side outlets and the same cross-sectional area. The two cylindrical pieces meet along a mitre. Development transfers the mitre-intersection heights to corresponding generators.

1. Draw the elevation of the two equal-diameter cylindrical pieces forming a 90° elbow and establish the symmetric mitre line.

2. Draw the circular end view and divide the half/whole circle into equal parts as required. Project these divisions as generators into the elevation.
3. Mark where each generator meets the mitre line.
4. Draw a stretch-out equal to πD, divide it into the same generator sequence, and erect generators.
5. Transfer the corresponding heights from the elevation to the stretch-out. Join the transferred points smoothly to form the cut edge.
6. Use symmetry to obtain the mating piece where appropriate; verify that both cut edges match.



Schematic process for the two-piece elbow. Exact geometry follows the dimensions given in the drawing problem.

Development errors: transferring apparent instead of true lengths; numbering generators differently in views and development; measuring around a circle with inconsistent chord steps; adding seam allowance when not asked; or reversing the mitre sequence.

Malayalam support: Elbow development-□ circle divisions elevation-□□□□□□ project □□□□□□ mitre line intersection heights □□□□□□□□□□□□□□. □□□ generator numbering stretch-out-□ □□□□□□□□□□ heights transfer □□□□□□□. Guess □□□□□□ curve □□□□□□□□□□□.

Module II summary and self-check

One-look summary

- Rectangular and concentric-circle methods are the prescribed ellipse methods.
- The prescribed parabola method is tangent method.
- An involute comes from an unwinding taut string.
- Prism/cylinder developments use parallel-line logic.
- Elbow development depends on generator-by-generator ordinate transfer.

Self-check

1. State two differences between ellipse and parabola.
2. Why must matching divisions be used in the concentric-circle method?
3. Draw an involute of a circle divided into 12 parts.
4. Find πD for D = 64 mm.
5. Explain why a mitre curve cannot be guessed.

Check for Exercise 4: π × 64 ≈ **201.1 mm**.

MODULE III · 9 INSTRUCTIONAL HOURS

Basic projections — theory, points and lines

CO3 **Understand** **3 + 3 + 3 h**

Projection converts spatial position into coordinated two-dimensional views. The safest method is not memorising isolated pictures; it is translating every phrase—above, below, in front, behind, parallel, perpendicular and inclined—into its effect on HP and VP.

Malayalam support: Point/line projection □□□□□□□□□□□ □□□□□ sentence split □□□□□□□: HP-□□□□ position □□□□□? VP-□□□□ position □□□□□? Front view VP-□□□ Top view HP-□□□ □□□□□□□□. XY line-□□□□ □□□ side-□□□□ □□□ view □□□□□□□□ □□□□□ quadrant rule □□□□□□□□□□□□□.

3.1 Theory of projection — first-angle and third-angle — 3 h

Reference elements

Vertical Plane (VP)

Receives the front view or elevation. It records height and width information.

Horizontal Plane (HP)

Receives the top view or plan. It records width and depth information.

XY reference line

Represents the intersection of VP and HP after HP is rotated into the drawing plane.

Projector

An imaginary line from object to projection plane. In orthographic projection it is perpendicular to the plane.

Front view

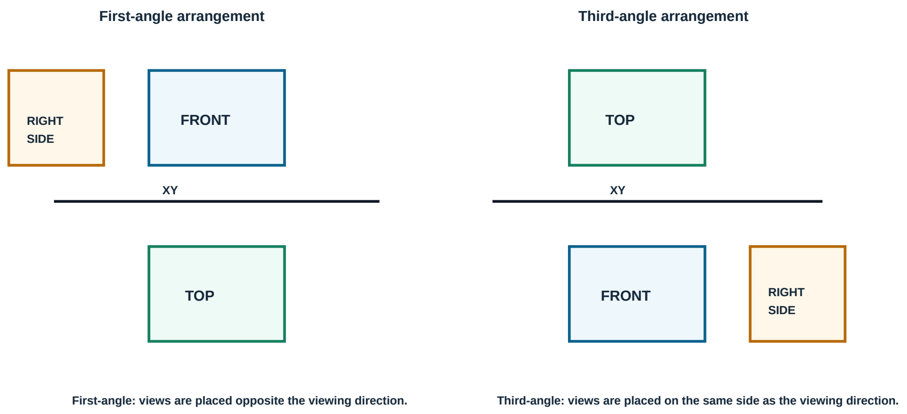
The projection on VP, conventionally labelled with a prime such as a′.

Top view

The projection on HP, conventionally labelled without a prime such as a.

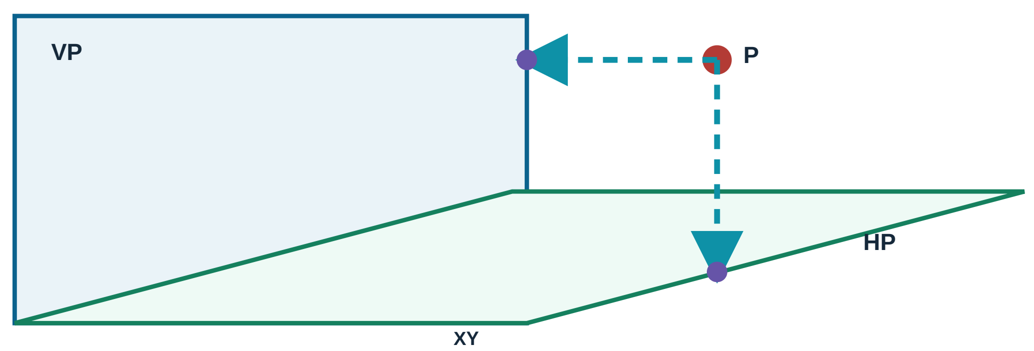
First-angle versus third-angle

Feature	First-angle projection	Third-angle projection
Relative location	Object is conceptually between observer and projection plane.	Projection plane is conceptually between observer and object.
Top view placement	Below front view.	Above front view.
Right-side view placement	To the left of front view.	To the right of front view.
Syllabus application	Official self-learning activity specifies first-angle method for constructing orthographic views.	Required for theory/identification comparison.



Arrangement comparison. The Module IV drawing exercises use first-angle projection as stated in the official self-learning activity.

Projection-ray visualisation



Static meaning: each view is formed where perpendicular projectors meet its projection plane.

Do not confuse view placement with viewing direction. The right-side view is obtained by looking from the right, but in first-angle projection it is placed on the left of the front view.

3.2 Projections of points in all four quadrants — 3 h

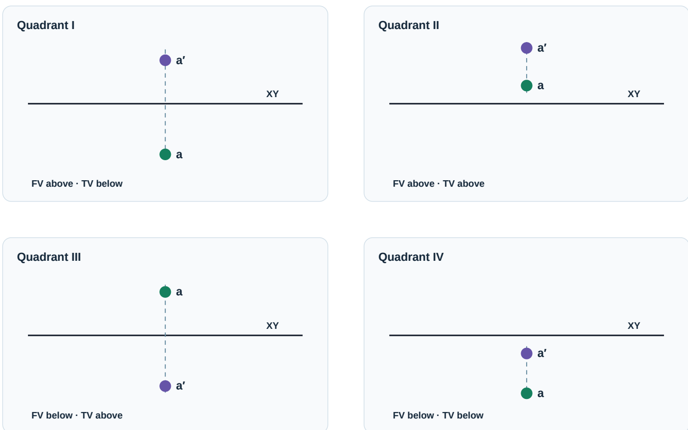
How a point creates two distances

Let point A be located at a stated distance from HP and VP. Its front view a′ is positioned from XY by the point’s distance above or below HP. Its top view a is positioned from XY by the point’s distance in front of or behind VP.

Distance of front view a′ from XY = distance of point from HP
Distance of top view a from XY = distance of point from VP

Quadrant sign-and-position chart

Quadrant	Physical position	Front view a′	Top view a
I	Above HP, in front of VP	Above XY	Below XY
II	Above HP, behind VP	Above XY	Above XY
III	Below HP, behind VP	Below XY	Above XY
IV	Below HP, in front of VP	Below XY	Below XY



Point-view locations after HP is rotated. a′ denotes front view; a denotes top view.

Demonstration A — first quadrant

Given: A is 25 mm above HP and 35 mm in front of VP.

1. Draw XY and one thin projector.
2. Mark a′ 25 mm above XY.
3. Mark a 35 mm below XY on the same projector.

Both views are aligned; their separation is 60 mm.

Demonstration B — third quadrant

Given: B is 20 mm below HP and 30 mm behind VP.

1. Draw XY and one thin projector.
2. Mark b′ 20 mm below XY.
3. Mark b 30 mm above XY.

Third quadrant: front view below, top view above.

Verbal decoding routine

1. Underline the HP phrase: above or below.
2. Underline the VP phrase: in front or behind.
3. Use HP phrase to place front view.
4. Use VP phrase to place top view.
5. Place both views on one projector and label correctly.

Malayalam support: “Above/Below HP” front view a′-□□□□ XY position □□□□□□□□□□□□□□□□□□. “In front of/Behind VP” top view a-□□□□ XY position □□□□□□□□□□□□□□□□□□. □ □□□□□ rules separate □□□ apply □□□□□□□ quadrant confusion □□□□□□.

3.3 Projections of lines — first quadrant, inclined to one plane only — 3 h

The official restriction is important: questions may be limited to lines in the **first quadrant** and inclined to **any one plane only**. This means the other plane relationship is normally parallel or perpendicular as stated.

True length and apparent length

A straight line appears in true length in a view when the line is parallel to that view’s projection plane. When inclined to the plane, its projection is shorter than true length. When perpendicular to the plane, its projection collapses to a point.

Standard line cases in the prescribed scope			
Spatial relation	**Front view on VP**	**Top view on HP**	**Key recognition**
Parallel to HP and VP	True length, parallel to XY	True length, parallel to XY	Both views equal TL
Perpendicular to HP; parallel to VP	True length, perpendicular to XY	Point	Top view collapses
Perpendicular to VP; parallel to HP	Point	True length, perpendicular to XY	Front view collapses
Parallel to VP; inclined θ to HP	True length at θ to XY	Shorter length, parallel to XY	Inclination visible in front view
Parallel to HP; inclined φ to VP	Shorter length, parallel to XY	True length at φ to XY	Inclination visible in top view

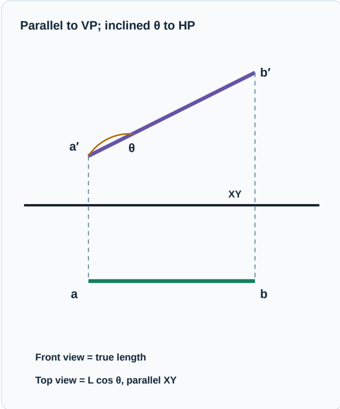
If a line of true length L is inclined by θ to HP and parallel to VP:
front-view length = L; top-view length = $L \cos \theta$.
If inclined by φ to VP and parallel to HP:
top-view length = L; front-view length = $L \cos \varphi$.

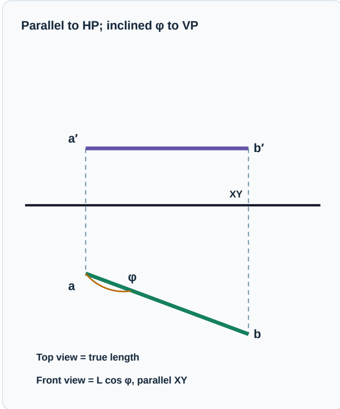
Construction: line parallel to VP and inclined to HP

1. Locate one end A from its distances above HP and in front of VP: a′ above XY and a below XY.
2. From a′ draw front view a′b′ equal to true length L at the given angle θ to XY.
3. Project b′ downward. Through a draw a line parallel to XY. Their intersection gives b.
4. Join ab. It is shorter than L and remains parallel to XY.

Construction: line parallel to HP and inclined to VP

1. Locate end A.
2. From a draw top view ab equal to true length L at angle φ to XY.
3. Project b upward. Through a′ draw a line parallel to XY. Their intersection gives b′.
4. Join a′b′. It is shorter than L and parallel to XY.





Within the syllabus scope, only one plane inclination is introduced at a time. The true-length view carries the given angle.

Numerical check — inclined to HP

A 70 mm line is parallel to VP and inclined 30° to HP.

Front view = 70 mm. Top view = 70 cos 30° = 60.62 mm.

Use the geometric projection for the drawing; the calculation is a verification.

Numerical check — inclined to VP

An 80 mm line is parallel to HP and inclined 45° to VP.

Top view = 80 mm. Front view = 80 cos 45° = 56.57 mm.

The apparent view cannot exceed the true length.

Common mistakes: drawing the angle in the wrong view; using apparent length as true length; placing first-quadrant top views above XY; misaligning end projectors; and forgetting that a line perpendicular to a plane projects as a point on that plane.

Malayalam support: Line □□□ plane-□□□□ parallel □□□, □ plane-□ true length □□□□□. HP-□□□□ angle □□□□□□□□□□□ line VP-□□□□ parallel □□□□□□□□ angle front view-□ □□□□□. VP-□□□□ angle □□□□□□□, line HP-□□□□ parallel □□□□□□□ angle top view-□ □□□□□.

Module III summary and self-check

One-look summary

- Front view forms on VP; top view forms on HP.
- First-angle top view is below front view.
- Point location is decoded separately from HP and VP phrases.
- A line parallel to a plane shows true length on that plane.
- A line perpendicular to a plane appears as a point on that plane.

Self-check

1. Place a point 18 mm above HP and 22 mm behind VP.
2. Place a point 25 mm below HP and 30 mm in front of VP.

3. Sketch both views of a line parallel to both planes.
4. For L = 60 mm, θ = 60°, calculate L cos θ.
5. Explain first-angle right-side view placement.

Checks: Exercise 1—both views above XY; Exercise 2—both below XY; Exercise 4—**30 mm**.

MODULE IV · 15 INSTRUCTIONAL HOURS

Projections–3D — isometric, orthographic and sectional views

CO4 **Understand / Apply** **3 + 6 + 6 h**

A single pictorial view helps visualisation, but manufacturing normally requires coordinated orthographic views. Sectioning then removes visual obstruction so internal shapes can be read without excessive hidden lines.

Malayalam support: Isometric view object-□□□□ 3D appearance □□□□□□□□□□□□. Orthographic projection shape-□□ Front, Top, Side views □□□ □□□□□□□□□□□□ exact information □□□□□□□□□. Sectional view imaginary cutting plane □□□□□□□□□□ inside details clear □□□□□□□□□□.

4.1 Orthographic and isometric projection concepts — 3 h

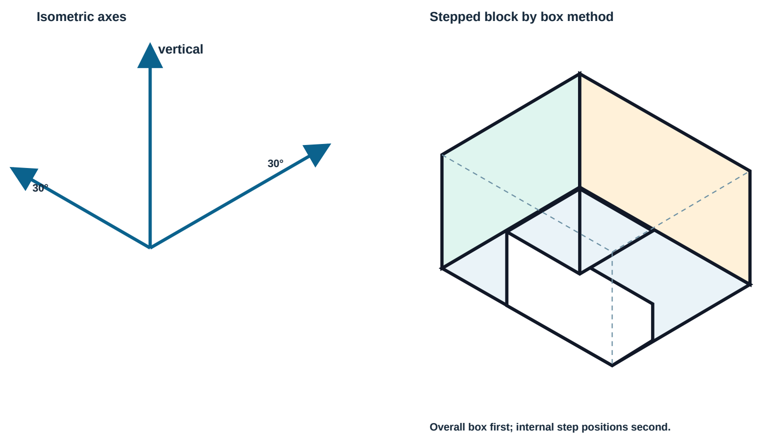
Pictorial versus multiview representation

Aspect	Isometric / pictorial view	Orthographic views
Main purpose	Quick spatial understanding	Accurate description of separate faces
Axes / projectors	Three principal axes appear 120° apart; two are commonly drawn 30° to horizontal	Projectors are perpendicular to projection planes
Circles	Generally appear as ellipses on isometric planes	Appear as circles when viewed normal to their plane
Dimensions	Useful for visualisation; must follow specified isometric rules	Primary production information is usually dimensioned on suitable true-shape views
View count	One pictorial view may show three faces	Front, top and side views are correlated

Isometric axes and box method

1. Choose a convenient lower corner as origin and draw one vertical axis plus two receding axes at 30° to the horizontal.
2. Construct an enclosing isometric box using the object’s overall length, width and height.

3. Locate changes of shape by measuring along isometric axes and drawing lines parallel to the axes.
4. Remove construction-box lines not required and darken visible final edges.



Isometric box method. All construction directions remain parallel to one of the three principal axes.

Choosing a front view

The best front view usually shows the characteristic shape, functional features and maximum detail with minimum hidden lines. The viewing direction must remain fixed once selected. A “front” is not automatically the longest face; it is the most informative face for the drawing purpose.

Isometric warning: non-isometric lines cannot be measured directly along an isometric axis. Locate their endpoints from controlling coordinates, then join them.

4.2 Orthographic projections of simple engineering objects — 6 h

First-angle arrangement used in the exercises

The official self-learning activity specifies the **first-angle projection method only** for constructing orthographic views from pictorial objects. Therefore: top view is placed below the front view; right-side view is placed to the left; left-side view is placed to the right.

View correlation

Front ↔ Top

Share the same width. Their corresponding points lie on common vertical projectors.

Front ↔ Side

Share the same height. Horizontal projectors transfer heights directly.

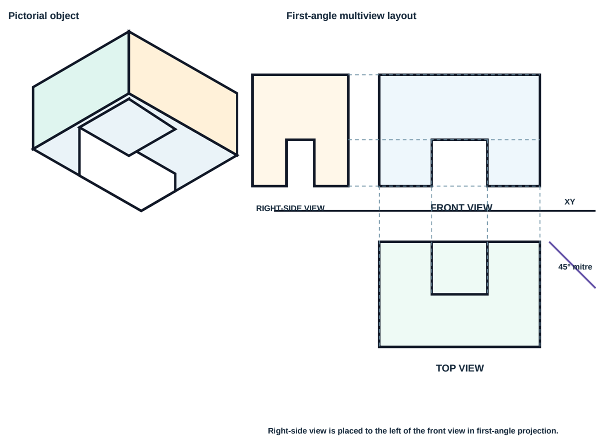
Top ↔ Side

Share the same depth. A 45° mitre line can transfer depth between plan and side view.

Width: common to front and top views
Height: common to front and side views
Depth: common to top and side views

General drawing sequence

1. Study the pictorial object. Mark overall width, height and depth; identify holes, slots, steps, slopes and symmetry.
2. Select the front view and reserve space for first-angle top and side views.
3. Draw XY and light overall bounding rectangles for each view.
4. Complete the front view first, then project width to the top view and height to the side view.
5. Transfer depth from top to side using a 45° mitre construction or approved direct method.
6. Add visible lines, necessary hidden lines and centre lines in proper priority.
7. Check alignment, count features in every view, then darken and label.



Coordinated first-angle views. The example is schematic; construct exactly from the dimensions supplied in the examination problem.

Features across views

Feature in object	What to check in views	Frequent mistake
Through hole	Circle in the normal view; hidden/visible parallel edges and centre line in other views	Omitting the centre line or showing wrong hole width
Step	Change of outline in views where visible; aligned break position	Different step location in front and top
Slot	Open or closed outline depending on viewing direction; centre/symmetry where relevant	Treating an open slot as a blind recess
Inclined face	May appear as an inclined edge in one view and a foreshortened area in another	Assuming every projected edge remains at the same angle
Curved surface	Tangent boundaries and centre lines must correlate	Adding a line where two smooth tangent surfaces meet without a real edge

Demonstration — view count check

A block has one central through hole and one top step. Before darkening, make a feature checklist:

1. Overall width appears equal in front and top views.
2. Overall height appears equal in front and side views.
3. Overall depth appears equal in top and side views.
4. The hole centre is aligned in all related views.
5. The hole is a circle only in the view normal to its axis.
6. The step break occurs at the same controlling coordinate in every relevant view.

This checklist catches more errors than repeatedly tracing the final outline.

Malayalam support: Front–Top views-□ width common, Front–Side views-□ height common, Top–Side views-□ depth common. □□□ feature-□□□ □□□□□ views-□□□ corresponding position □□□□□ □□□□□ check □□□□□□□.

4.3 Full sectional views of simple engineering objects — 6 h

Why sectioning is used

Hidden lines can make an internal shape difficult to read. In a sectional view, an imaginary cutting plane passes through the object, the material in front is conceptually removed, and the remaining part is viewed. The syllabus/self-learning scope is **full sectional views only**.

Cutting-plane line

Indicates where the object is cut and includes arrows showing viewing direction. Ends are labelled, commonly A–A.

Section lining / hatching

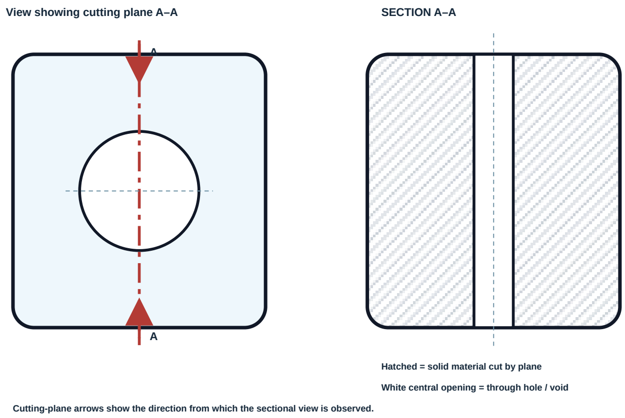
Thin, evenly spaced lines show solid material actually cut by the plane. A convenient angle such as 45° is commonly used.

Void

Holes, slots and empty spaces are not hatched. Their boundary remains visible in the section.

Full-section procedure

1. Select a cutting plane that passes completely through the important internal feature.
2. Show the cutting-plane position and arrow direction in the adjacent view.
3. Project the sectional view as if the portion between observer and cutting plane were removed.
4. Draw the remaining visible outline and the boundaries of holes/voids.
5. Apply thin, uniform section lines only to cut solid material.
6. Remove hidden lines made unnecessary by the section, unless a hidden feature remains essential under the adopted convention.
7. Add centre lines and label the section, for example SECTION A–A.



Full section through a central hole. The hole remains unhatched; only the cut material receives section lines.

Section-line quality rules

- Use thin continuous lines with uniform spacing.
- Keep the angle consistent within one component.
- Stop hatching exactly at boundaries; do not pass through holes or outside the object.
- For adjacent separate parts, change angle or spacing so parts remain distinguishable.

- Do not use thick object-line weight for hatching.
- Do not hatch empty spaces, cavities or holes.

What the cutting plane does not do

The cutting plane is imaginary; it does not physically shorten the object or change dimensions. Features behind the cutting plane remain visible if they are in the viewing direction. A section is not simply “fill the whole view with diagonal lines.”

Demonstration — central through hole

1. Place A–A through the axis of the hole.
2. Project the outer profile.
3. Show the hole as an unhatched opening.
4. Hatch solid material on both sides uniformly.
5. Retain the centre line.

The section immediately communicates the hole shape without hidden-line clutter.

Demonstration — stepped recess

1. Pass the full cutting plane through the recess centre.
2. Project each depth change at its true coordinate.
3. Leave the recess space unhatched.
4. Hatch only cut walls and base material.
5. Check that the section agrees with the adjacent view.

A recess boundary may be visible even though the empty recess itself is never hatched.

Common mistakes: hatching the hole; forgetting cutting-plane arrows; using hidden lines everywhere even though the section reveals the feature; changing hatch spacing randomly; and drawing a half section when the official scope asks for full sectional views.

Malayalam support: Cutting plane touch □□□□□□□□□ solid material □□□□□□ hatch □□□□□□□. Hole, slot, cavity □□□□□□□ empty space hatch □□□□□□□□□. Full section-□ cutting plane □□□□□□ object-□□□□□ □□□□□□□□□□; half section □ syllabus activity-□□□□ scope □□□□.

Module IV summary and self-check

One-look summary

- Isometric gives a compact 3D impression.
- Orthographic views preserve coordinated width, height and depth.
- Course practice uses first-angle arrangement.
- Full section removes the front half conceptually.
- Hatch solid material cut by the plane—not voids.

Self-check

1. Where is the first-angle top view placed?
2. Which dimensions are shared by top and side views?
3. Why is a 45° mitre line used?

4. Draw a full section of a block with one central hole.
5. List four hatching errors.

Checks: top view below front; top and side share depth; mitre transfers depth.

FORMULA / RULE / PROCEDURE BANK

High-value rules for drawing work

Scale relationships

RF = drawing length / actual length
drawing length = RF × actual length

Both lengths must be in the same unit before division.

Regular polygons

Interior angle = $(n-2)180°/n$
Exterior angle = $360°/n$

Calculation supports checking; instrument construction controls the sheet.

Cylinder development

Stretch-out = πD
Lateral area = πDh

Add end circles for the official closed-cylinder exercise.

Prism development

Stretch-out = base perimeter
Lateral area = perimeter × height

Preserve side sequence and attach both ends for a closed prism.

Line projection check

Apparent length = L cos(inclination)

Use only for the stated one-plane inclination cases; projected length cannot exceed true length.

View correlation

Front ↔ Top: width
Front ↔ Side: height
Top ↔ Side: depth

Use projectors and a 45° mitre line, not visual guessing.

Procedure bank

Task	Start with	Control step	Verification
Dimensioning	Functional geometry and unit convention	Place extension and dimension lines clearly	No duplicate or missing dimensions
Curve construction	Axes / base geometry	Number equal divisions consistently	Symmetry and smoothness
Involute	Base curve and unwinding direction	Cumulative length on successive tangents/arcs	Curve starts tangentially and grows smoothly
Development	Correct views and generator numbering	Transfer true lengths/ordinates	Mating edges and perimeter agree
Point projection	XY and one projector	Decode HP and VP phrases separately	Quadrant table matches view positions
Line projection	One end and the true-length view	Place given angle in the correct view	Other view is aligned and not longer than TL
Orthographic views	Front view and overall boxes	Project width/height/depth	Every feature appears consistently
Full section	Cutting-plane location and viewing arrows	Draw remaining outline and unhatched voids	Only cut solid is hatched

Drawing-sheet quality and safety bank

Instrument safety

Handle compass and divider points carefully; close or cap them when not in use. Do not leave sharp instruments projecting beyond the board.

Ergonomics

Use a stable board, adequate lighting and a posture that avoids prolonged neck and wrist strain. Take short visual breaks during long sheets.

Sheet quality

Keep hands and instruments clean, avoid repeated erasing that damages the surface, and secure the sheet without distortion.

DIAGRAM AND VISUAL LIBRARY

One-look visual recall

Line language

Thick continuous: visible outline
Dashed: hidden detail
Long–short chain: centre/axis
Thin continuous: construction, projection, dimension, hatching

Review Module I diagram

Conic framework

Ellipse: two perpendicular axes and central symmetry.
Parabola: one axis, one vertex, open curve.
Never connect curve points with straight segments.

Review conic diagrams

Projection signs

FV distance ↔ HP relation.
TV distance ↔ VP relation.
First quadrant: FV above XY, TV below XY.

Review quadrant chart

First-angle layout

Top below front.
Right-side view left of front.
Width, height and depth must correlate.

Review orthographic diagram

Development cue

Prism: panel strip.
Cylinder: πD rectangle.
Elbow: generator ordinates transferred from mitre.

Review developments

Section cue

Arrows define viewing direction.
Cut solid = hatched.
Void = unhatched.
Centre line retained.

Review full section

True-length cue

Parallel to plane → true length on that plane.
Perpendicular to plane → point on that plane.

Review line cases

Polygon cue

Given side is the datum. Use a suitable approved construction and verify every side with a divider.

Review polygon table

SOLVED EXAMPLES AND DEMONSTRATIONS

Step-by-step checks across all four modules

1 · Reducing-scale calculation

Problem: Represent 7.5 m at RF 1:25.

Given: actual length = 7.5 m = 7500 mm; RF = 1/25.

Required: drawing length.

Substitution: (1/25) × 7500 = 300 mm.

Final: 300 mm. Check: 300 × 25 = 7500 mm.

2 · Polygon angle verification

Problem: verify the interior angle of a regular octagon.

n = 8. Interior angle = (8−2)×180°/8 = 1080°/8 = 135°.

Final: 135°. Exterior angle check: 360°/8 = 45°; 135° + 45° = 180°.

3 · Cylinder stretch-out

Problem: D = 72 mm, h = 95 mm.

Stretch-out = πD = 226.19 mm. Lateral area = πDh = 21,488.5 mm^2.

Final: lateral rectangle ≈ 226.2 mm × 95 mm; add two Ø72 mm ends for the closed exercise.

4 · Point in second quadrant

Problem: P is 32 mm above HP and 18 mm behind VP.

1. p′ = 32 mm above XY.
2. p = 18 mm above XY.
3. Both lie on one projector.

Interpretation: both views above XY identifies the second quadrant.

5 · Line inclined to HP

Problem: L = 90 mm, parallel VP, θ = 40° to HP.

Front view = true length = 90 mm at 40° to XY. Top-view check = 90 cos 40° ≈ 68.94 mm, parallel XY.

Final: 90 mm FV; ≈68.9 mm TV. Geometric construction remains the submitted method.

6 · Orthographic correlation audit

Problem: an object is 80 wide, 50 high and 40 deep.

- Front bounding box: 80 × 50.
- Top bounding box: 80 × 40.
- Side bounding box: 40 × 50.

Check: width 80 repeats in front/top; height 50 in front/side; depth 40 in top/side.

PRACTICAL / ACTIVITY CENTRE

Official 15-week self-learning plan — 22.5 hours

Each listed week carries 1.5 hours in the official syllabus. Use instructor-approved dimensions, sheet format and construction method. The activity record below is a study/record aid, not a substitute for department instructions.

Suggested student activities for self-learning				
Week	**Module**	**Official activity coverage**	**Hours**	**Suggested evidence**
1	I	Concepts of drawing; materials/instruments; sheet sizes; board types; sheet layout, title block and types of lines	1.5	Instrument list + line/sample sheet
2	I	Correct dimensioning standards; redraw and dimension a given figure as per BIS; Plain/Diagonal/Vernier scales for study	1.5	Corrected dimensioned drawing + scale notes
3	I	Regular polygon definition/terminology; construct pentagon, hexagon, heptagon, octagon with side given	1.5	One verified polygon sheet
4	II	Conic terminology; ellipse by rectangular and concentric-circle methods; parabola by tangent method	1.5	Three prescribed constructions
5	II	Introduction to miscellaneous curves; involute of circle and rectangle	1.5	Two involute constructions
6	II	Development procedure for applications; closed rectangular prism, closed cylinder, 90° two-piece symmetric elbow with standard side outlets	1.5	Three labelled developments
7	II	Series exam preparation	1.5	Timed mixed drawing practice
8	III	Theory of projection; first-angle and third-angle projection	1.5	Comparison chart and symbols
9	III	Projection of points in first, second, third and fourth quadrants	1.5	Four-quadrant problem sheet
10	III	Lines in first quadrant: parallel to both; parallel to one and perpendicular to other; parallel to one and inclined to other	1.5	Line-case sheet with TL checks
11	IV	Orthographic and isometric projection; simple pictorial isometric drawings	1.5	Two pictorial objects
12	IV	Front, top and side views; orthographic projections of simple objects from pictorial views, first-angle only	1.5	One complete three-view sheet
13	IV	Sectional-view familiarisation; full sectional views of simple objects	1.5	One cutting-plane and section sheet
14	IV	Series exam preparation	1.5	Timed two-module drawing set
15	All	Revision of all topics	1.5	Error log and final checklist
Total self-learning hours			**22.5**	**15 weekly records**

Standard activity procedure

Before drawing

1. Write activity title, date, module and CO.
2. Copy only instructor-approved problem data.
3. Select sheet orientation and reserve title-block space.

4. Inspect instruments and sharpen pencils safely.
5. Plan view locations before drawing any final line.

After drawing

1. Check dimensions/side lengths with scale or divider.
2. Check view alignment and labels.
3. Remove only distracting construction remnants; do not damage the sheet.
4. Record errors and corrections.
5. Obtain instructor verification where required.

Reusable drawing record format

Activity / sheet number:	**Date / module / CO:**
Aim:	**Given data and method:**
Instrument / sheet setup:	**Construction sequence:**
Verification performed:	**Result / inference:**
Errors noticed and correction:	**Instructor signature / remarks:**

Values and equipment: exact problem dimensions, drawing-sheet size, pencil grades and institutional submission format are not specified in the supplied syllabus. Use the current instructor/laboratory/drawing-hall instruction rather than inventing values.

EXPECTED / PRACTICE QUESTIONS

Module-wise question bank with model-answer points

These are instructional expected/practice questions prepared from official syllabus topics. They are not claimed as previous or official SITTTR questions.

Module I — Introduction

Very Short Answer

Q1. What is the purpose of a centre line?

Show answer

A centre line indicates an axis, centre of a circle/arc, or line of symmetry. It is drawn as a thin chain line and normally extends slightly beyond the feature.

Short Answer

Q2. Differentiate object, hidden and construction lines.

Show answer

- **Object line:** thick continuous; visible edges/outlines.
- **Hidden line:** thin/medium dashed; invisible edges.
- **Construction line:** very thin continuous; temporary locating geometry.

Descriptive

Q3. State six principles of good dimensioning.

Show answer

Any six: dimension each feature once; avoid missing dimensions; keep dimensions outside views when possible; avoid crossing; do not dimension to hidden lines when a clear view exists; use correct Ø/R/angle symbols; use one unit convention; use uniform arrowheads/text; dimension from functional references; never require measurement from the picture.

Numerical Practice

Q4. Calculate the drawing length of 18 m at RF 1:60.

Show answer

18 m = 18,000 mm. Drawing length = (1/60)×18,000 = **300 mm**.

Drawing Procedure

Q5. Explain how you would construct and verify a regular hexagon when its side is given.

Show answer

1. Draw the given side AB.
2. Use a standard hexagon construction; a regular hexagon has circumradius equal to side.
3. Locate the circle centre and draw the circumcircle.
4. Step the side chord six times around the circle.
5. Join consecutive vertices.

6. Verify every side with the divider before darkening.

Application

Q6. A drawing is attractive but has repeated dimensions and equal line weights. Is it acceptable?

Show answer

No. Repeated dimensions create conflict risk and equal line weights destroy line-function hierarchy. Technical clarity and unambiguous interpretation take priority over decoration.

Module II — Basic Figures

Very Short Answer

Q7. Name the prescribed ellipse methods.

Show answer

Rectangular method and concentric-circle method only.

Short Answer

Q8. Why must both concentric circles use the same radial divisions?

Show answer

Each ellipse point combines coordinates from matching angular positions on the major-radius and minor-radius circles. Mismatched divisions combine unrelated positions and produce a distorted curve.

Drawing Procedure

Q9. Write the construction sequence for an ellipse by concentric-circle method.

Show answer

1. Draw major/minor axes.
2. Draw concentric circles with radii equal to semi-major and semi-minor axes.
3. Divide both by common radial lines.
4. Draw verticals from outer-circle points and horizontals from matching inner-circle points.
5. Mark intersections.
6. Join smoothly through axis ends.

Concept

Q10. Define involute and state two applications.

Show answer

An involute is the locus of the end of a taut string unwound from a base curve. Applications include involute gear-tooth profiles and geometrical analysis of winding/unwinding or cam-type paths.

Numerical Practice

Q11. Find the lateral development size of a Ø84 mm cylinder 120 mm high.

Show answer

Stretch-out = $\pi D = \pi \times 84 = 263.89$ mm. Height = 120 mm. The lateral rectangle is approximately **263.9 mm × 120 mm**. For the closed exercise add two Ø84 mm end circles.

Descriptive

Q12. Explain development of a two-piece symmetric 90° elbow.

Show answer

1. Draw equal-diameter elbow elevation and symmetric mitre.
2. Divide circular end view and project generators.
3. Find generator intersections with mitre.
4. Stretch out πD with identical division sequence.
5. Transfer each mitre ordinate to the matching generator.
6. Join transferred points smoothly.
7. Check matching edge/symmetry of the second piece.

Module III — Basic Projections

Very Short Answer

Q13. Where is the top view placed in first-angle projection?

Show answer

Below the front view.

Comparison

Q14. Compare first-angle and third-angle view placement.

Show answer

First-angle places top below front and right-side view left of front. Third-angle places top above front and right-side view right of front. Their conceptual object/plane positions are also reversed.

Point Projection

Q15. A point is 24 mm below HP and 36 mm behind VP. Place its views.

Show answer

The point is in the third quadrant. Draw one projector: front view 24 mm below XY; top view 36 mm above XY.

Point Projection

Q16. Both views of a point are below XY. Which quadrant is possible?

Show answer

Fourth quadrant: the point is below HP and in front of VP.

Line Projection

Q17. Describe the views of a line perpendicular to HP and parallel to VP.

Show answer

Its front view on VP is the true length and is perpendicular to XY. Its top view on HP collapses to a point.

Numerical Practice

Q18. A 100 mm line is parallel to HP and inclined 60° to VP. Find the projected lengths.

Show answer

Top view = true length = 100 mm at 60° to XY. Front view = 100 cos60° = **50 mm**, parallel to XY.

Module IV — Projections-3D

Very Short Answer

Q19. What is shared between front and side views?

Show answer

Height.

Short Answer

Q20. Why is a 45° mitre line useful?

Show answer

It transfers depth measurements between the top view and side view while maintaining view correlation.

Comparison

Q21. Differentiate isometric and orthographic representation.

Show answer

Isometric gives one pictorial 3D impression along three axes; orthographic gives separate 2D views using perpendicular projectors and is preferred for exact face information. Circles may appear elliptical in isometric but circular in a normal orthographic view.

Procedure

Q22. List the sequence for drawing three first-angle views from a pictorial object.

Show answer

1. Study features and choose front.
2. Plan first-angle layout.
3. Draw bounding boxes.
4. Complete front view.
5. Project top view below.
6. Transfer height/depth to side view, using mitre if needed.
7. Add visible, hidden and centre lines.
8. Check all feature correlations and darken.

Sectional View

Q23. What is hatched in a full section?

Show answer

Only solid material actually intersected by the cutting plane. Holes, slots and cavities remain unhatched.

Troubleshooting

Q24. A section shows the hole filled with hatching and no cutting-plane arrows. Identify the faults.

Show answer

- The hole is a void and must not be hatched.
- Cutting-plane arrows and labels are required to identify viewing direction and section reference.

- Centre line should be retained for the hole.

EXAMINATION PRACTICE

Practice Model Paper — not an official SITTTR model question paper

This practice set follows the official ESE pattern stated in the supplied syllabus: 2.5 hours, 60 marks, 15 marks per module; Part A contains two 5-mark choices per module and Part B contains one 10-mark set with choice per module. The actual examination questions and dimensions are set by the competent authority.

Part A — answer one of the two questions from each module (4 × 5 = 20)

Module I

1A. Redraw a simple plate supplied by the instructor and apply correct dimensioning conventions. *OR*

1B. Construct a regular octagon when its side is given and show construction lines.

Module II

2A. Construct an ellipse by the concentric-circle method for the given axes. *OR*

2B. Draw the involute of a circle of given diameter.

Module III

3A. Draw the projections of four points, one in each quadrant, from stated distances. *OR*

3B. Draw projections of a line parallel to VP and inclined to HP, with one end located in the first quadrant.

Module IV

4A. Draw the isometric view of a simple stepped block from supplied dimensions. *OR*

4B. Draw the front and top views of a simple pictorial object in first-angle projection.

Part B — answer one set with choice from each module (4 × 10 = 40)

Module I — 10

5A. Prepare a standards sheet showing line types, title-block concept, and correct dimensioning of a supplied component. *OR*

5B. Construct the prescribed pentagon, hexagon, heptagon and octagon for instructor-supplied side lengths.

Module II — 10

6A. Develop the complete surface of a closed cylinder of supplied diameter and height. *OR*

6B. Develop one piece of a two-piece symmetric 90° elbow from supplied views.

Module III — 10

7A. Solve a combined set of point projections in four quadrants with complete labels. *OR*

7B. Draw two prescribed first-quadrant line cases: one perpendicular to a plane and one inclined to one plane only.

Module IV — 10

8A. Draw front, top and right-side views of a supplied object by first-angle method. *OR*

8B. Draw a full sectional front view and corresponding top view for a supplied object and cutting plane.

Practice-paper evaluation guide

Use the following non-official self-check priorities:

- **Construction method:** correct geometric sequence and reference data.
- **Accuracy:** dimensions, side equality, smooth curves, view alignment.
- **Standards:** line types, labels, symbols, hatching and first-angle placement.
- **Completion:** all requested views/end faces/curves included.
- **Presentation:** readable, clean and logically spaced—without erasing essential construction evidence.

Official internal examination pattern

The official internal drawing examination is 2 hours and covers any two modules. For each selected two-module block, Part A shows 3 questions totalling 15 marks (5 each), and Part B shows 2 sets with choice totalling 10 marks; the complete paper totals 50 raw marks.

QUICK REVISION

Exam-ready one-look review

Module I

- Line meaning and hierarchy
- Title block and sheet layout
- No repeated dimensions
- RF in common units
- Four regular polygons from side

Module II

- Ellipse: two prescribed methods
- Parabola: tangent method
- Involute: cumulative unwinding
- Prism/cylinder: parallel-line development
- Elbow: transfer generator ordinates

Module III

- FV on VP; TV on HP
- First-angle top below front
- Four-quadrant position table
- Parallel to plane = true length
- Perpendicular to plane = point

Module IV

- Isometric axes and box method
- Width/height/depth correlation
- First-angle three-view layout
- 45° mitre transfers depth
- Full section: hatch cut solid only

Common mistakes to eliminate

- Darkening construction before verification
- Using one line weight everywhere
- Duplicating dimensions
- Mixing units in RF
- Joining curve points with straight chords
- Reversing involute direction
- Changing generator numbering
- Using apparent length in development
- Putting first-quadrant top view above XY
- Drawing line inclination in the wrong view
- Placing right-side view on the right in first-angle
- Breaking feature alignment across views
- Hatching a hole or cavity
- Omitting cutting-plane arrows
- Applying a half section when full section is asked

Final sheet-submission checklist

Geometry

All stated sizes used; sides/arcs/axes verified; curves smooth; true lengths transferred correctly.

Standards

Correct line types, labels, symbols, view arrangement, dimensioning and section lining.

Presentation

Views fit the sheet, title block is complete, no clipped dimensions, no excessive erasing, and final outlines are legible.

GLOSSARY

Essential technical terms

Term	Concise meaning	Malayalam support
Axis	A reference line about which geometry is symmetric or rotates.	symmetry/rotation □□□□□□□□□□□□□□□□□ reference line.
Dimension	A numerical specification of size or location.	size □□□□□□□□□□□ position □□□□□□□□□□□□□□□□□ numerical value.
Extension line	Thin line extending from a feature to a dimension line.	feature-□ □□□□□□□ dimension line □□□ □□□□□□□□ thin line.
Representative Fraction	Drawing length divided by actual length in the same units.	same unit-□□□□□ drawing/actual length ratio.
Regular polygon	Polygon with all sides and all interior angles equal.	□□□□□ sides-□□ angles-□□ equal □□ polygon.
Major axis	Longest diameter of an ellipse.	Ellipse-□□□□ longer central axis.
Minor axis	Shortest diameter of an ellipse, perpendicular to major axis.	major axis-□□□□ perpendicular □□ shorter axis.
Tangent	Line touching a curve locally without crossing it at that point.	curve-□□ local □□□ touch □□□□□□□□□□ line.
Involute	Path of the end of a taut string unwound from a base curve.	tight string unwind □□□□□□□□□□□□ end trace □□□□□□□□□□ path.
Generator	A line element that generates a ruled surface such as a cylinder.	surface form □□□□□□□□ □□□□□□□□□□□□□ longitudinal line element.
Development	Flat pattern of a three-dimensional surface.	3D surface flat sheet □□□ □□□□□□□ pattern.
Projector	Imaginary line used to transfer object points to a projection plane.	object point-□□ projection plane-□□□□□□□ transfer □□□□□□□□□□ line.
HP	Horizontal Plane receiving the top view.	Top view □□□□□□□□□□□ Horizontal Plane.
VP	Vertical Plane receiving the front view.	Front view □□□□□□□□□□□ Vertical Plane.
XY line	Reference line representing intersection of HP and VP after rotation.	HP–VP intersection represent □□□□□□□□□□□ reference line.
True length	Actual line length visible when the line is parallel to the projection plane.	plane-□□□□ parallel □□□□□□□□ view-□ □□□□□□□□ actual length.
Orthographic projection	Multiview projection using projectors perpendicular to planes.	perpendicular projectors □□□□□□□□□□□□□□□ multiview method.
Isometric view	Pictorial view along three equally inclined principal directions.	three principal directions □□□□□□□□□□□□□□□ pictorial view.
Cutting plane	Imaginary plane used to expose internal details.	inside detail □□□□□□□□□□□ □□□□□□□□□□□□□□□□ imaginary plane.
Section lining	Thin parallel lines showing solid material cut by a plane.	cut □□□□□ solid material □□□□□□□□□□□□□□□□□□ hatch lines.

Textbooks, references and web resources

Textbooks listed in the supplied syllabus

No.	Title	Author	Publication
1	Engineering Graphics	P I Varghese	VIP Publishers
2	Engineering Graphics	K. C John	PHI Learning Private Limited
3	Engineering Graphics unique methods easy solutions	K N Anilkumar	Adhyuth Narayan Publishers

References listed in the supplied syllabus

No.	Title	Author	Publication
1	Engineering Drawing	N. D. Bhatt	Charotar Publishing House, Anand, Gujarat
2	Engineering Graphics with AutoCAD	D.M. Kulkarni, A.P. Rastogi, A.K. Sarkar	PHI Learning Private Limited
3	Engineering Drawing	M. B. Shah and B.C.Rana	Pearson Publications

Web resources listed in the supplied syllabus

1.
2.
3. external edits in>